

2008 JC2 PRELIMINARY EXAMINATIONS
H2 Biology
Paper 3 (Applications)
MARK SCHEME

Disclaimer

This mark scheme should be used purely as a revision tool.
 Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

Key to mark scheme

ref.	with reference to
/	alternative wording
(brackets)	recommended wording that is optional
<u>underline</u>	accept this word only
OVP	other valid points
AW	alternative wording

Section A

QUESTION 1 [Total: 15]

Babies born with severe combined immune deficiency (SCID) have no defence against common infections and quickly become ill when the protection from maternal antibodies is lost. This was the result of a mutation in their ADA gene that resulted in infants suffering a deficiency in all functions of their immune system.

(a) Explain briefly how a mutation in the ADA gene leads to “deficiency” in the functions of the immune system. [3]

- 1 mutation on chromosome 20;
- 2 genetic deficiency in / non-functional enzyme adenosine deaminase (ADA deficiency);
- 3 involve in purine metabolism / deamination / hydrolysis of / breaking down adenosine & deoxyadenosine;
- 4 toxic / deoxyATP byproducts accumulates;
- 5 harms lymphocytes in immune system by blocking lymphocyte replication;
- 6 prevents maturation & unable to produce lymphocytes / kills

Gene therapy for SCID has been carried out using the procedure shown in Fig. 1.1.

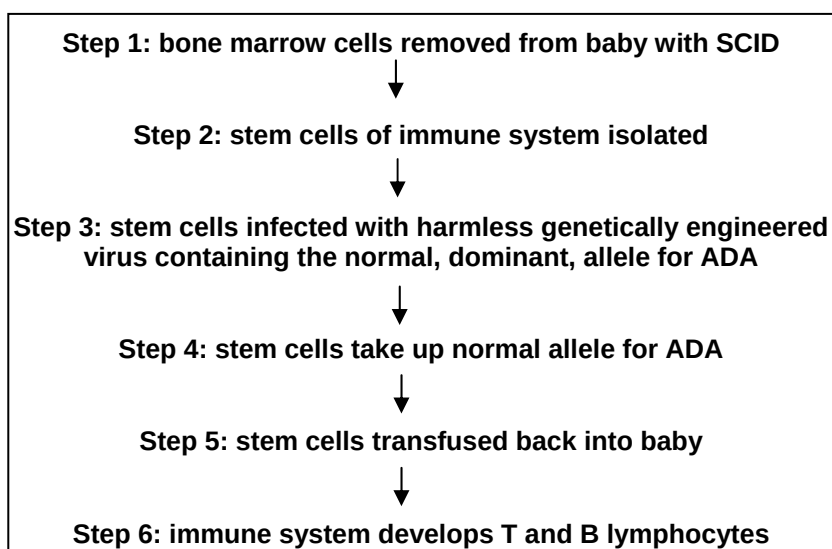


Fig. 1.1

(b) (i) Describe the properties of the stem cells used in Fig. 1.1. [2]

- 1 multipotent, capable of giving rise to specialized cells / cells of related functions (e.g. lymphocytes);
- 2 no tissue-specific structures that allow specialized functions, unspecialised;
- 3 capable of proliferation / long-term self-renewal / replicate themselves;

R: capable of mitosis to form more cells

(ii) List an advantages and disadvantage of using the human stem cells mentioned in Fig. 1.1. [2]Advantages

- 1 immune response is unlikely because patients are using their own cells;
- 2 some are easily available (e.g. blood stem lines);
- 3 partly specialized, requires less coaxing to create specialized cells;

Disadvantages

- 4 limited longevity / non immortal, difficult to maintain in cell cultures for long periods / limited no. of cell divisions;
- 5 only multipotent; limited flexibility; cannot become any cell type;
- 6 questionable quality due to genetic defects; targeted disease may still be present in stem cell lines;

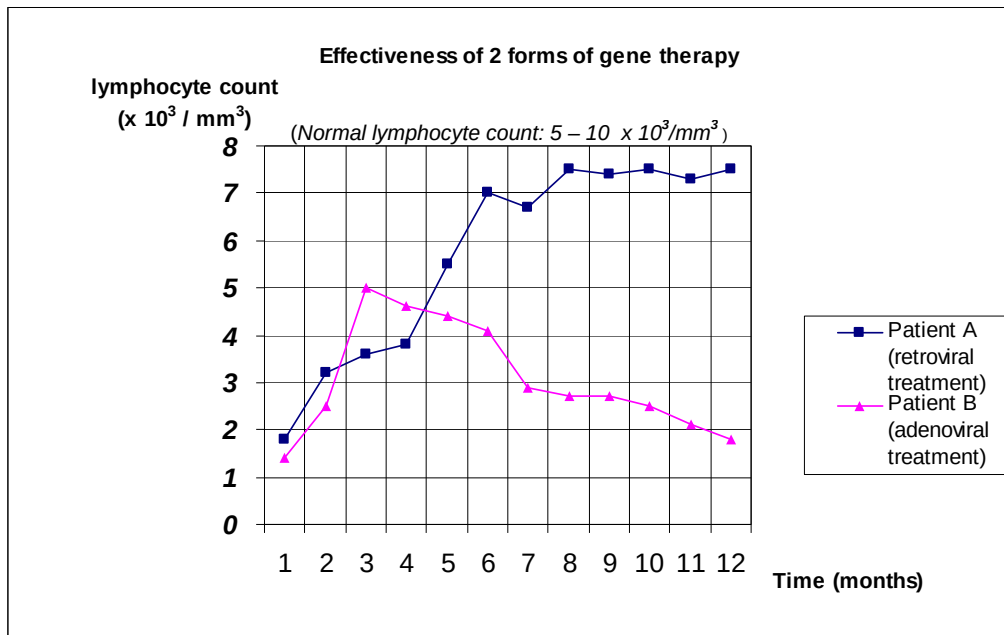
(c) (i) Explain what is meant by *gene therapy*. [2]

- 1 transfer/addition of one/more normal (or modified) allele / gene into an individual's body cells;
- 2 to correct a genetic defect (or boost resistance to a disease); / a form of somatic cell therapy;
- 3 via the use of vectors (viral and non-viral) for the insertion of a transgene / to restore normal phenotype / normal phenotype can be expressed

(ii) Why is it easier to perform gene therapy when the normal allele is the dominant allele of the gene concerned (step 3)? [2]

- 1 has immediate effect when added / inserted into genome;
- 2 only 1 normal allele needs to be inserted since it is dominant and not masked;
- 3 no need to remove / inactivate, recessive / mutant allele;

The diagram below shows the lymphocyte counts (both T & B) of 2 infant patients that underwent 2 separate types of gene therapy treatments for SCID. Patient A was treated with a retrovirus while Patient B was treated with an adenovirus. The healthy lymphocyte count of the patients was monitored by a doctor at the end of every month for a period of 12 months. (The acceptable range of lymphocyte count for non-SCID patients is $5-10 \times 10^3/\text{mm}^3$).



With reference to diagram,

(d) (i) Describe the trends observed in the 2 patients' lymphocyte count. [2]

- 1 patient A lymphocyte count increase from 1.8 to $7.5 \times 10^3 / \text{mm}^3$ & entered normal range by 5th month;
- 2 patient B lymphocyte count increase to $5 \times 10^3 / \text{mm}^3$, entered into normal range by 3rd month but decreased thereafter to $1.8 \times 10^3 / \text{mm}^3$;

*No mention of 'normal range' – max 1m

(ii) Account for the difference in lymphocyte count at the end of 12 months. [2]

- 1 retroviral vector allowed for integration of recombinant DNA that could be stably propagated hence resulting in long term stable expression / pass from parent to daughter cells;
- 2 adenovirus however could not integrate / exists extrachromosomally, into chromosomal DNA hence transferred gene does not segregate equally into daughter cells [R: DNA gets degraded]

QUESTION 2 [Total: 15]

A 5% herbicide solution of cyanamide kills the weeds which infest fields of tobacco plants, but causes some damages to the tobacco. However, tobacco plants have been genetically engineered by inserting a gene from a soil fungus, *Myrothecium verrucaria*. The gene codes for an enzyme, cyanamide hydratase, which converts cyanamide into urea.

The gene was inserted into a plant virus and then tobacco plants were infected with the virus. A strain of transgenic tobacco plants was thus developed that carried the gene for cyanamide hydratase but were unaffected by other viral proteins. The enzyme was most active in the leaves, showing about one fifth of the activity found in *Myrothecium*.

The transgenic plants thrive when treated with a dilute solution of cyanamide, since the plants have urease enzymes which convert the urea to ammonium ions which provide the plants with a source of nitrogen.

(a) Explain why the tobacco plants described in the passage are *transgenic plants*. [2]

- 1 acquired a foreign gene; / gene from another species; / recombinant DNA molecule from 2 different species;
- 2 fungus gene coding for cyanamide hydratase;
- 3 gene expressed where cyanamide hydratase breaks down cyanamide to urea;

(b) Why was the gene for cyanamide hydratase inserted into a virus? [1]

- 1 plants cannot take up foreign gene on their own;
- 2 virus as vector to introduce foreign DNA into host/plant;

(c) Describe briefly another way of transforming plant cells other than the method used in this experiment. [1]

- 1 High particle bombardment / "gene gun" method / microinjection;
- 2 electroporation;
- 3 infection with *Agrobacterium* / Ti Plasmid;

Equal masses of tobacco plant callus tissue were cultured for four weeks on media containing different concentrations of two plant growth regulators: auxin and cytokinin. The results are shown in Table 2.1 below.

treatment	concentration of plant growth regulators / mg dm ⁻³		effect of plant growth regulators on callus growth
	auxin	cytokinin	
A	2.00	0.00	little or no growth
B	2.00	0.02	growth of roots
C	2.00	0.20	increased growth of callus with no differentiation
D	2.00	0.50	growth of shoots
E	0.00	0.20	little or no growth

Table. 2.1

(d) Explain what is meant by *callus tissue*. [1]

- 1 Mass of disorganized / undifferentiated / unspecialized plant cells;

(e) With reference to Table 2.1,**(i) Describe the effects of auxin and cytokinin on callus growth. [4]**

- 1 both cytokinin and auxin needed for cell division / growth / little or no growth if only one is present
- 2 lowest conc. of cytokinin cf. auxin gives only roots / AW;
- 3 highest conc. of cytokinin cf. auxin gives only shoots / AW;
- 4 intermediate cytokinin to auxin concentration, gives more callus / stimulates, cell division / growth;
- 5 appropriate use of figures with units; / pts 2-4 support with values
- 6 ref. to different concentrations (i.e. auxin conc. always higher throughout);

(ii) Suggest what treatment should be given to the shoots developed from callus in treatment D to turn them into plantlets. [2]

- 1 remove (from treatment D);
- 2 give treatment B / description of treatment B;
- 3 justification; / need to develop roots in order to develop plant;

Subsequently, extensive field trials were carried out to assess the environmental effects of growing these transgenic tobacco plants.

(f) (i) Give 3 reasons why it is important to assess the environmental effects of growing these transgenic tobacco plants. [3]

- 1 possibility of transfer / cross-pollination of introduced herbicide resistance gene from transgenic plants to wild relatives;
- 2 development of superweeds that are difficult to eradicate; / plant establish itself as a weed elsewhere;
- 3 need to assess possible effects of transgene products / enzymes / proteins / toxins on other organisms (e.g. insects / endangered wild species) that feed on the tobacco plant;
- 4 assess possible effects on the ecosystem (i.e., upset ecological balance / threaten biodiversity);

(ii) Other than providing resistance to herbicide, explain another way in which genetic engineering of tobacco plants could increase crop yield.

- 1 improve resistance to pest;
- 2 resistance to herbicide;
- 3 resistance to diseases (e.g. tobacco mosaic viruses);
- 4 fast maturation; / code for protein that results in faster growth;
- 5 better / healthier growth;

QUESTION 3 [Total: 15]

RFLP analysis has facilitated the process of DNA fingerprinting and disease detection played an important role in the Human Genome Project.

One of its application, DNA fingerprinting, has been widely used to distinguish between individuals using only samples of their DNA. Forensic scientists can make DNA fingerprints from materials such as blood, skin and semen to identify criminals.

(a) Outline how genetic fingerprinting is carried out. [4]

- 1 DNA is extracted from cell sample and cut by restriction enzyme(s);
- 2 Enzymes chosen to cut close to but not within VNTRs;
- 3 Fragments separated by gel electrophoresis according to their molecular size;
- 4 Transferred to a membrane by Southern blotting;
- 5 Single-stranded probes which hybridise / are complementary the DNA sequence / VNTR;
- 6 Radioactive probes are detected using autoradiography / use of X-ray film;

(b) Explain why VNTRs are useful genetic makers for DNA fingerprinting whereas genes are seldom used for DNA fingerprinting. [4]

- 1 Variable No. of Tandem Repeats / lengths of repeated base sequence; (**R**: large no. of varieties)
- 2 in DNA that does not code for a gene;
- 3 number / size of repeats differs markedly between individuals; / highly polymorphic repeats / VNTRs;
- 4 may exist in abundance throughout the genome;
- 5 only identical twins have the same, number of repeats / genetic fingerprint;
- 6 inherited, half from each parent;
- 7 VNTRs can be probed for sequence flanking it;

Max 2

- 8 genes encodes a functional polypeptide;
- 9 sequence of DNA must be conserved due to its important biological function;
- 10 mutation rate lower than non-coding regions;
- 11 not able to differentiate between individuals;
- 12 gene sequence need to be known before probing / probe is made;

Max 2

(c) There are still genetic diseases for which no gene has yet been discovered. Until the gene can be located, cloned, and sequenced, no probe can be made to detect it directly. However, alleles for Huntington's disease and a number of other genetic diseases could be detected in an indirect way using RFLP. Explain how this can be done. [4]

- 1 a closely linked RFLP marker / disease gene is located very close to the RFLP;
- 2 crossing over between the marker and the gene is very unlikely to occur during gamete formation;
- 3 the marker and gene will almost always be inherited together;
- 4 presence of the RFLP can serve as a surrogate/indicator for the disease gene itself;
- 5 even though the RFLP marker is not part of the gene;

(d) How did RFLPs play an important role in the Human Genome Project? [3]

- 1 scattered throughout the genome / genome wide
- 2 served as genetic markers
- 3 use for making genome wide linkage map / find out the order of RFLP markers on a chromosome
- 4 ref. frequency of crossing over between 2 RFLP markers to indicate the distance between two loci on a chromosome

Section B**QUESTION 4 [Total: 20]****(a) Distinguish between a genomic and cDNA library. [6]****[1 mark per point]**

Genomic library	cDNA library
1 collection of all the cloned <u>DNA fragments</u> from the <u>entire genome</u>	represents the subset of genes / obtained from reverse transcription of mature mRNA in the starting cell
2 consists of large amount of DNA	consists of much smaller amount of DNA
3 presence of introns	absence of introns
4 useful if unsure as to what cell type the gene is expressed	5 study genes responsible for the specialized functions of a particular cell type
6 useful if unable to obtain the particular cell type	7 trace changes in patterns of gene expression during development / make cDNA from cells of the same type at different times in the life of an organism.
8 to analyse regulatory sequences or the introns associated with the gene	9 know the coding sequences of the gene / to deduce the amino acid sequence of the protein from the DNA sequence

(b) Outline the procedures for the cloning of a named eukaryotic gene using a plasmid. [8]**[1 mark per point]****Preparation of insert and vector**

- 1 digest genomic DNA from human cells;
/ *ref.* make copy of gene from mRNA using reverse transcriptase;
- 2 digest a suitable vector / plasmid with same restriction enzyme used;

Formation of recombinant DNA molecules

- 3 mixing restriction fragments / fragments anneal via complementary sticky ends;
- 4 incubation with DNA ligase to produce a recombinant plasmid;

Transformation

- 5 transformation / introduce recombinant plasmid into host / bacterial cells;
- 6 use competent bacterial cells / presence of calcium ions with heat-shock treatment / electroporation;

Identification of bacterial colonies

- 7 select cells that have taken up recombinant plasmid / insulin gene;
- 8 *ref.* antibiotic resistance gene and β -galactosidase gene;
- 9 grow cells in culture medium with antibiotics and X-gal;
- 10 select bacterial colonies that are resistant to antibiotic;
- 11 inserted gene disrupts β -galactosidase gene hence colonies appear white (in colour);
- 12 *ref.* to blue colour of X-gal after digestion
- 13 *ref.* replica plating (onto nitrocellulose filter);
/radioactive/fluorescent tagged;
/ gene probe to verify presence of gene;
- 14 *ref.* gene probe being a single-stranded DNA strand complementary to insulin gene
/ hybridise to insulin gene;
- 15 localisation of position by autoradiography

Large-scale culture of bacteria

- 16 grow bacteria containing insulin gene in fermenters / large quantities to clone the gene;

Alternative: (PCR description) – max 5 marks

- 1 heat to 95°C to denature / H-bonds broken, to separate DNA strand;
- 2 synthesise primers that flank the 3' ends of target region;
- 3 cooling to 54°C to facilitate annealing of primers;
- 4 heating to 72°C to facilitate elongation;
- 5 addition of Taq / DNA polymerase to synthesise complementary strand;
- 6 starting at position of primers / oligonucleotides;

(c) Discuss the difficult ethical concerns of the human genome project.**[6]****[1 mark per point]****1) Privacy and confidentiality of genetic information**

- difficult to determine who owns and controls genetic information;

2) Fairness in the use of genetic information

- by insurance companies, employers, courts, schools, adoption agencies, and the military;
- difficult to determine who should have access to personal genetic information and how it will be used;

3) Commercialization of products

- including property rights (patents, copyrights, and trade secrets) and accessibility of data and materials;
- difficult to determine who owns genes and other pieces of DNA;
- difficult to ascertain if patenting DNA sequences will limit their accessibility and development into useful products;

4) Fairness in access to advanced genomic technologies.

- difficult to ascertain who will benefit or if it will result in major worldwide inequities;

5) Clinical issues

- including the education of doctors and other health-service providers, people identified with genetic conditions, and the general public about capabilities, limitations, and social risks; and implementation of standards and quality-control measures;
- problems associated with the evaluation and regulation for accuracy, reliability and utility of genetic tests;
- the need to prepare healthcare professionals for the new genetics as well as the public to make informed choices;

6) Uncertainties associated with gene tests for susceptibilities and complex conditions

- e.g., heart disease, diabetes, and Alzheimer's disease – questions related to whether:
 - tests should be performed when no treatment is available;
 - parents have the rights to have their minor children tested for adult-onset diseases;
 - genetic tests are reliable and interpretable by the medical community;

7) Reproductive issues

- including adequate and informed consent and use of genetic information in reproductive decision making;
- possibility of improper counselling of parents by healthcare personnel about the risks and limitations of genetic technology;
- difficult to determine the reliability and usefulness of fetal genetic tests;

8) Psychological impact, stigmatization, and discrimination

- due to an individual's genetic differences;
- possibility that personal genetic information as perceived by society, may affect an individual in various ways;

9) Conceptual and philosophical implications

- regarding human responsibility, free will vs genetic determinism, and concepts of health and disease;
- some of the related perceptions and uncertainties include:
 - whether behaviour is determined by genes;
 - what is the acceptable level of diversity;
 - whether there is a clear distinction between medical treatment and enhancement;

- *headings must accompany some discussion*